

Phase 4: Component Selection

Marklin WiFi AC Train Controller

2026-03-04

Test Project: Component Selection

Design Constraints

- First revision uses an ESP32-C3-MINI-1 module (not bare chip). This simplifies BOM - no crystal, flash IC, or RF matching network needed.
- Components shall be SMT as far as possible; passives in 0603 packages.

Block 1: Power Supply

Bridge Rectifier

Parameter	Selected	Alternative	Rejected
Part	MB6S	DB107S	KBP206
Manufacturer	ST	Hottech	--
LCSC	(Semtech) C2978785	--	Too large
Package	SOP-4	SOP-4	(through-hole) Through-hole
Rating	(MBS) 600V, 0.5A	1000V, 1A	--
Price	~\$0.024	~\$0.03	--

Selected: MB6S - SMD package, 600V rating adequate for 24 VAC (peak ~34V), 0.5A sufficient (MCU domain only, motor current does not pass through PSU).

Buck Converter

Parameter	Selected	Alternative 1	Alternative 2
Part	XL1509-3.3E	LM2596S-3.3	RT7272BGS
Manufacturer	XLSEMI	UMW (TI)	P. Richtek
LCSC	C74193	(clone) C347420	C127866
Package	SOP-8	TO-263-5	SOIC-8-EP
Vin range	4.5-40V	4.5-40V	4.5-36V
Iout max	2A	3A	3A
Frequency	150 kHz	150 kHz	500 kHz
Efficiency	~85%	~85%	~95%
Price	\$0.14	\$0.24	\$0.26
Stock	42,500+	5,200+	117,000+

Selected: XL1509-3.3E1 - Lowest cost, smallest package, 40V max input covers 24 VAC rectified (34V peak) with margin. 2A output is 4x the 500 mA requirement. Simple application circuit. Well-proven in hobbyist designs.

Why not RT7272BGSP? Better efficiency (95% vs 85%) and synchronous rectification, but overkill for 300 mA typical load. Adjustable output adds complexity. Worth considering if thermal issues arise in enclosed locomotive body.

Why not LM2596S-3.3? Excellent documentation but TO-263-5 package is physically large. Better suited for bench supplies than compact train controllers.

Buck Converter Support Components

Component	Value	Package	LCSC	Notes
Input cap	100uF/50V	8x10mm	C299392	Smoothing after bridge rectifier
Input ceramic cap	electrolytic 100nF/50V	0805	⁶ C49678	High-frequency bypass
Output cap	220uF/10V	6.3x7.7mm	C289139	Output ripple reduction
Output ceramic cap	electrolytic 22uF/10V	0805	⁹ C566252	Low-ESR output filtering
Schottky diode	SS34 (3A/40V)	SMA	³ C8678	Catch diode for XL1509
Inductor	68uH/2A	8x8mm	C408345	Per XL1509 datasheet for 3.3V
Feedback divider	Per datasheet	0805	--	output Fixed output version, internal divider

Input EMI Filter (optional, recommended)

Component	Value	Package	LCS	Notes
Common mode choke	10mH/0.5A	Through-hole	--	To be selected in detailed design
X-cap	100nF/275VAC	--	--	Across AC input

Block 2: ESP32-C3 MCU

MCU Module

Parameter	Selected	Alternative
Part	ESP32-C3-MINI-1-N	ESP32-C3-MINI-1U-N
Manufacturer	⁴ Espressif	⁴ Espressif
LCSC	C2838502	C2911374
Antenna	PCB (integrated)	IPEX (external)
Flash	4 MB	4 MB
Price	\$2.01	\$2.14
Stock	9,700+	2,300+

Selected: ESP32-C3-MINI-1-N4 - PCB antenna version. Lowest cost, highest stock. External antenna (1U variant) only needed if WiFi range is insufficient inside the locomotive body - unlikely for home layout distances.

Decoupling Capacitors

Component	Value	Qty	Package	LCSC	Notes
Bulk decoupling	10uF/10V ceramic	2	0805	C1585	VDD3P3 and
HF decoupling	100nF/25V ceramic	4	0603	⁰ C1466	VDD3P3 RTC Each VDD pin

Boot/Reset

Component	Value	Packa	LCSC	Notes
Reset button	3x4mm tactile	SMD	C288662	Active-low reset (EN pin)
Boot button	3x4mm tactile	SMD	C288662	GPIO9 low = download
Pull-up resistor (EN)	10K	0603	C25804	mode EN pin pull-up
Pull-up resistor	10K	0603	C25804	Boot pin pull-up (run mode)
(GPIO9) Reset cap	100nF	0603	C14663	EN pin RC delay

Block 3: Zero-Crossing Detector

Optocoupler

Parameter	Selected	Alternative	Rejected
Part	EL357N(C)(TA)-G	PC817C	6N137
Manufacturer	Everlight	Sharp	--
LCSC	C29981	C66340	Overkill
Package	SOP-4	DIP-4	speed --
CTR	130-260% (C)	200-400% (C)	--
Speed	rank) ~4us rise	rank) ~4us rise	--
Isolation	3.75kV	5kV	--
Price	~\$0.03	~\$0.04	--

Selected: EL357N(C)(TA)-G - SMD package, adequate speed for 50/60 Hz zero-crossing detection (~20ms period, <100us timing requirement easily met). High CTR C-rank gives clean output transitions.

Why not PC817? DIP package is larger. Both work, but EL357N is SMD and slightly cheaper.

Support Components

Component	Value	Package	Notes
AC series resistor	3.3K/1W	1206 or	Limits LED current to ~10mA at 34V peak
AC series resistor	3.3K/1W	axial 1206 or	Two in series for voltage sharing (6.6K total)
DC pull-up resistor	10K	axial 0603	Pull-up to 3.3V on output (C25804)
Anti-parallel diodes	1N4148 x2	SOD-323	Protect optocoupler LED from reverse voltage

Block 4: Hardware Interlock

Logic IC

Parameter	Selected	Alternative
Part	74LVC00AD,11	SN74LVC00APW

Manufacturer	Nexperia	Texas
LCSC	C6044	Instruments C7803
Package	SOIC-14	TSSOP-14
VCC range	1.65-5.5V	1.65-5.5V
Price	\$0.13	\$0.10
Stock	Good	Good

Selected: SN74LVC00APWR (TI) - Slightly cheaper, TSSOP-14 saves board space. 3.3V operation, 5ns propagation delay (vastly faster than needed). Quad NAND - uses 2 gates for cross-coupled interlock, 2 gates spare.

Support Components

Component	Value	Package	Notes
Decoupling cap	100nF/10V	0603	VCC bypass (C14663)

Block 5: TRIAC Drive (x2)

TRIAC Gate Drive Optocoupler

Parameter	Selected	Alternative
Part	MOC3021S-TA1	MOC3021
Manufacturer	Lite-On	(DIP) Lite-On
LCSC	C131706	C123202
Package	SMD-6P	DIP-6
Type	Random-phase	Random-phase
Isolation	(non-zero-cross) 5.3kV	5.3kV
Price	\$0.14	\$0.086
Stock	Good	30,000+
Qty needed	2	--

Selected: MOC3021S-TA1 - SMD version of the MOC3021, per design requirement for SMT components. Random-phase type required for phase-angle motor speed control (must fire at arbitrary points in AC cycle). Zero-cross types (MOC3041) would only allow on/off control, defeating the purpose.

Power TRIAC

Parameter	Selected	Alternative	Rejected
Part	BT136-600E	BTA16-600B	BTA08-600
Manufacturer	WEIDA	Minos	C
LCSC	C2901007	C5452761	--
Package	TO-252 (SMD)	TO-220	--
Rating	600V / 4A	(THT) 600V / 16A	--
Gate trigger	~10mA	~50mA	--
Price	(sensitive) \$0.11	\$0.35	--
Stock	1,820	370	--

Qty needed	2	--	--
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Selected: BT136-600E - 4A rating provides 2x margin over typical motor current (<2A). Sensitive gate (10mA) is easier to drive from MOC3021 output. SMD package (TO-252) saves space. Significantly cheaper than BTA16-600B.

Why not BTA16-600B? 16A is massive overkill for a model train motor. More expensive, lower stock, requires through-hole mounting. The BT136 at 4A is adequate with margin.

Support Components (per channel, x2)

Component	Value	Package	Notes
Optocoupler LED	470R	0603	Drive current from 3.3V: $(3.3-1.2)/470 = \sim 4.5\text{mA}$
Gate resistor	360R/0.5W	1206	Between MOC3021 output and TRIAC gate
Snubber resistor	100R/0.5W	1206	Across TRIAC, in series with snubber cap
Snubber capacitor	100nF/400V	Film/ceramic	Across TRIAC, dV/dt protection

Block 6: Voltage Sensing

Optocoupler

Same as zero-crossing detector: EL357N(C)(TA)-G (C29981). Keeps BOM simple.

Support Components

Component	Value	Package	Notes
AC series resistor 1	10K/0.5W	1206	Current limiting into optocoupler LED
AC series resistor 2	10K/0.5W	1206	Voltage sharing
DC load resistor	10K	0603	Sets DC output level (C25804)
Filter cap	1uF/10V	0603	Low-pass filter (~16 Hz with 10K)
ADC protection resistor	1K	0603	Series with ADC input (C21190)
Clamp diode to 3V3	BAT54S	SOT-23	Schottky clamp, overvoltage protection

Block 7: Status LED

Component	Value	Package	LCSC	Notes
LED (green)	0805	0805	C2297	Standard green, 20mcd typ
Current limiting resistor	SMD 1K	0603	C2119	~1.5mA at 3.3V (adequate brightness for indicator)

Block 8: USB Programming

USB-C Connector

Parameter	Selected	Alternative
Part	TYPE-C 16P	TYPE-C 6P
Manufacturer	SHOU HAN	SHOU
LCSC	C2906290	HAN
Pins	16 (full)	6
Price	\$0.044	(simplified) \$0.020

Selected: 16-pin USB-C - Full pinout for proper USB 2.0 operation. At \$0.044 the cost difference from 6-pin is negligible.

ESD Protection

Parameter	Selected
Part	USBLC6-2SC6
Manufacturer	TECH PUBLIC (or ST)
LCSC	C2827693 (TP) or C7519
Package	(ST) SOT-23-6
ESD rating	±30kV air, ±25kV contact
Capacitance	0.4pF (minimal signal)
Price	Impact \$0.02 (TP) / \$0.07 (ST)

Selected: USBLC6-2SC6 - Industry standard for USB ESD protection. TECH PUBLIC clone acceptable for prototype; use ST for production.

Support Components

Component	Value	Package	Notes
CC1	5.1K	0603	USB-C CC1 pin to GND (device mode)
pull-down CC2	5.1K	0603	(C23186) USB-C CC2 pin to GND (device mode)
pull-down			(C23186)

Block 9: Connectors

Track Input (2-pin)

Parameter	Selected
Part	5.08mm 2-pin screw
Manufacturer	terminal XINLAIYA
LCSC	XYEEK500-5.08-2P
Rating	300V / 10A
Price	\$0.098

Motor Output (3-pin)

Parameter	Selected
Part	5.08mm 3-pin screw
Manufactur	terminal DORABO
LCSC	DB2EV-5.08-3P-GN
Rating	300V / 10A
Price	\$0.050

Note: 5.08mm pitch screw terminals provide adequate current rating (10A) and are easy to wire in the field. Pluggable variants allow disconnecting the locomotive without tools.

Common Passives (shared across blocks)

Component	Value	Packag	LCSC	Usage
Resistor 10K	10K 1%	0603	C2580	Pull-ups, dividers
Resistor 1K	1K 1%	0603	C2119	LED, series
Resistor 5.1K	5.1K 1%	0603	C2318	protection USB CC
Resistor 470R	470R 1%	0603	C2317	Optocoupler LED
Cap	100nF	0603	C1466	drive Decoupling
100nF/25V Cap 10uF/10V	X7R 10uF X5R	0805	C1585	Bulk decoupling
Cap 1uF/10V	1uF X7R	0603	C1584	Filtering
Diode 1N4148	75V/150m	SOD-32	C2128	Signal/protection

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BOM Summary

Block	Key Active Components	Estimated Cost	Active
1. Power Supply	MB6S + XL1509 + SS34 +	~\$0.50	
2. ESP32-C3 MCU	inductor ESP32-C3-MINI-1-N4	~\$2.01	
3. Zero-Crossing	EL357N	~\$0.03	
4. HW Interlock	SN74LVC00APWR	~\$0.10	
5. TRIAC Drive (x2)	MOC3021S x2 + BT136-600E x2	~\$0.50	
6. Voltage Sense	EL357N + BAT54S	~\$0.06	
7. Status LED	LED + resistor	~\$0.01	
8. USB Programming	USB-C + USBLC6-2SC6	~\$0.07	
9. Connectors	Screw terminals (2+3 pin)	~\$0.15	
Total	active	~\$3.43	

components

Passives (resistors, caps, diodes) add approximately \$0.30-0.50, bringing the estimated total BOM cost to ~\$3.90-4.10 (at LCSC single-unit pricing, excluding PCB).

Components Reviewed and Rejected

Component	Reason for Rejection
ESP32-C3 bare chip	Requires external crystal, flash, RF matching - too complex for Rev 1
ESP32-C3-MINI-1U (IPEX)	External antenna unnecessary for home layout range
LM2596S-3.3 (buck)	TO-263 package too large for in-locomotive PCB
RT7272BGSP (buck)	Higher efficiency but overkill; adjustable adds complexity
BTA16-600B (TRIAC)	16A massively oversized for <2A motor; more expensive, through-hole
MOC3041 (zero-cross opto)	Zero-cross type prevents phase-angle speed control
PC817 (optocoupler)	DIP package larger than EL357N SMD; otherwise equivalent
6N137 (optocoupler)	High-speed opto - overkill for 50/60 Hz zero-crossing
MOC3021 DIP-6	Design requires SMT components; replaced by MOC3021S-TA1
(through-hole) 0402 passives	(SMD-6P) Design specifies 0603 package for passives
6-pin USB-C connector	Marginal cost savings (\$0.02), incomplete USB-C implementation

Open Items for Detailed Design

1. XL1509 application circuit - verify inductor value and Schottky diode rating against datasheet reference design
2. Snubber values - 100nF + 100R are starting values; may need tuning based on TRIAC dV/dt behavior with inductive (motor) load
3. EMI filter - common mode choke and X-cap values to be determined during layout/testing
4. Zero-crossing resistor values - verify 2x 3.3K (6.6K total) gives adequate LED current across full AC cycle for reliable detection
5. Thermal analysis - BT136-600E in TO-252 package at 2A needs ~1.6W dissipation check; may need copper pour for heat sinking
6. LCSC stock verification - confirm all parts in stock at time of ordering (stock levels from March 2026 research)